

NAAN MUDHALVAN MINI PROJECT

Implement Client Script & UI Policy (Incident)

PROJECT OVERVIEW

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Documentation Note: This project documentation has been adapted from an existing reference implementation of the Naan Mudhalven mini project. The technical structure, ServiceNow configuration, implementation approach, and testing workflow are retained as the reference basis, with team-specific information updated for the current team.

1. Introduction

Background

ServiceNow is a cloud-based platform widely used for IT Service Management (ITSM). One of its core processes is Incident Management, whose goal is to restore normal service operation as quickly as possible after an unplanned disruption. Every disruption is captured as an Incident record that carries fields such as Impact, Urgency, Priority, Assignment Group, Assigned To and State.

The quality of these fields decides how fast and how correctly an incident is handled. If a critical incident is saved without an owner, or if its urgency does not match its impact, response time and reporting accuracy both suffer.

Problem Statement

In a default ServiceNow instance, users are free to leave important fields empty or to change them in ways that do not follow the business process. For High Impact incidents in particular, the following problems are common:

- High Impact incidents are saved without an Assignment Group or an Assigned To user.
- Urgency is set inconsistently, so the calculated Priority does not reflect the real severity.
- State is changed directly from the Incident list, bypassing the checks available on the form.

Objectives

This project applies configuration-level controls to the Incident table to solve the problems above. The objectives are to:

- Enforce mandatory fields automatically when Impact is High.
- Keep Urgency consistent with Impact by setting it automatically and locking it.
- Validate the record before it is saved, and show clear messages to the user.
- Restrict risky direct edits of State from the list view while keeping form updates available.
- Return the form to its normal behaviour when Impact is no longer High.

Approach

The solution combines two ServiceNow client-side features. UI Policies provide declarative, no-code control of field behaviour (mandatory, read-only, visible). Client Scripts, written in JavaScript, handle logic that a UI Policy cannot express: onChange for automatic updates, onSubmit for validation, and onCellEdit for list editing.

Project at a Glance

Property	Value
Platform	ServiceNow
Target Table	Incident
Trigger Field	Impact
UI Policy	High Impact Control (2 UI Policy Actions)
Client Scripts	onChange, onSubmit, onCellEdit
Fields Controlled	Assignment Group, Assigned To, Urgency, State

2. Project Overview

Purpose

The purpose of this project is to improve data accuracy and consistency while creating and updating Incident records in ServiceNow.

The project uses UI Policies and Client Scripts to control Incident fields dynamically, automatically populate values, validate required information, and prevent incorrect updates. The system responds to changes in the Impact field and applies appropriate controls to fields such as Assignment Group, Assigned To, Urgency, and State.

Features

The major features of the project are:

1. Makes Assignment Group mandatory when Impact is High.
2. Makes Urgency read-only when Impact is High.
3. Automatically sets Urgency to High when Impact is changed to High.
4. Prevents saving a High Impact Incident when Assigned To is empty.
5. Prevents users from changing State directly from the Incident list.
6. Allows State changes through the Incident form.
7. Reverses UI Policy restrictions when Impact is changed from High to another value.

3. Architecture



Figure: High-level architecture of the Incident controls

User Interface

The project uses the ServiceNow Incident form and Incident list as its user interface. Users can create, view, and update Incident records through the ServiceNow platform. The interface contains fields such as:

- Impact
- Urgency
- Assignment Group
- Assigned To
- State

UI Policies dynamically control field behavior based on the values entered by the user.

Client-Side Logic

Client Scripts are used to control user interactions on the Incident form and list. The project implements three types of Client Scripts:

- **onChange** – automatically updates Urgency when Impact changes.
- **onSubmit** – validates Assigned To before saving the Incident.
- **onCellEdit** – prevents State from being changed directly from the Incident list.

Data Storage

Incident records are stored and managed by the ServiceNow platform. The project works with the Incident table and does not directly access the underlying database.